# CHAROTAR UNIVERSITY OF SCIENCE TECHNOLOGY

**Sixth Semester of B. Tech. (EC) Examination**

**December 2013**

**EC 311 VLSI TECHNOLOGY AND DESIGN**

**Date: 06.12.2013, Friday Time: 1:30 a.m. To 4:30 p.m. Maximum Marks: 70**

*Instructions:*

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

## SECTION –I

**Q-1 Answer the following questions. (any four)** **12**

(a) Explain impact of different VLSI design styles upon the design cycle time and the achievable circuit performance.
(b) What is a short channel effect? Explain in a brief.
(c) Discuss substrate bias effect for CMOS inverter.
(d) Derive the switching power dissipation of CMOS inverter.

(e) Explain the logic of pass transistor.
(f) Explain oxide related capacitance in brief.
(g) Explain Generalized NAND structure with Multiple inputs using combinational MOS logic circuits.

**Q-2(a)** Draw the band diagram of the nMOS structure at the surface inversion .Derive the expression of the threshold voltage for the same. **7**

(b) Measured voltage and current data for MOSFET are given below. Determine the type of the device and calculate the parameters $k_n$, $V_{TO}$, and $\gamma$. Assume Fermi potential is -0.3V **4**

| $V_{GS}$(V) | $V_{DS}$(V) | $V_{SB}$(V) | $I_D$(μA) |
|---|---|---|---|
| 3 | 3 | 0 | 97 |
| 4 | 4 | 0 | 235 |
| 5 | 5 | 0 | 433 |
| 3 | 3 | 3 | 59 |
| 4 | 4 | 3 | 173 |
| 5 | 5 | 3 | 347 |

**OR**

| | | |
|---|---|---|
| **Q-2(a)** | Compare the two technology scaling methods, namely (i) constant electric filed scaling and (ii) constant voltage scaling. In particular show analytically by using equations how the delay time, power dissipation and power density are affected in terms of the scaling factor S. | 5 |
| **(b)** | Explain circuit operation of CMOS inverter. Derive critical voltage points $V_{IL}$, $V_{IH}$, $V_{OH}$, $V_{OL}$ for the same. | 6 |
| **Q-3(a)** | Define the propagation delay $\tau_{PHL}$ and $\tau_{PLH}$. Derive $\tau_{PLH}$ for CMOS inverter. | 7 |
| **(b)** | Explain D latch circuit with its waveform. | 5 |

**OR**

| | | |
|---|---|---|
| **Q-3(a)** | Derive the threshold voltage expression for CMOS two input NAND Gate. | 6 |
| **(b)** | Consider the following Boolean function<br><br>$Z = \overline{A(D + E) + BC}$<br><br>Realize Boolean function using nMOS complex logic gates and complex CMOS logic gates. | 6 |

## SECTION –II

| | | |
|---|---|---|
| **Q-4 (a)** | Discuss basic steps for fabrication and briefly explain fabrication process of pMOS transistor. | 5 |
| **(b)** | Explain the concept of regularity, modularity and locality in brief. | 6 |
| **Q-5 (a)** | Explain voltage Bootstrapping in detail. | 6 |
| **(b)** | Explain working of resistive load nMOS inverter in brief. | 6 |

**OR**

| | | |
|---|---|---|
| **Q-5 (a)** | Explain Dynamic CMOS Logic (Precharge-Evaluate Logic) in detail. Also illustrate the cascading problem in dynamic CMOS logic. | 6 |
| **(b)** | Explain principal of sequential logic circuits with proper example. | 6 |
| **Q-6** | Write a short note on following ( any three) | **12** |
| **(a)** | CMOS ring oscillator circuit. | |
| **(b)** | VLSI design flow | |
| **(c)** | SRAM Vs DRAM | |
| **(d)** | Complimentary pass transistor logic (CPL) | |
| **(e)** | Depletion load nMOS inverter operation. | |

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